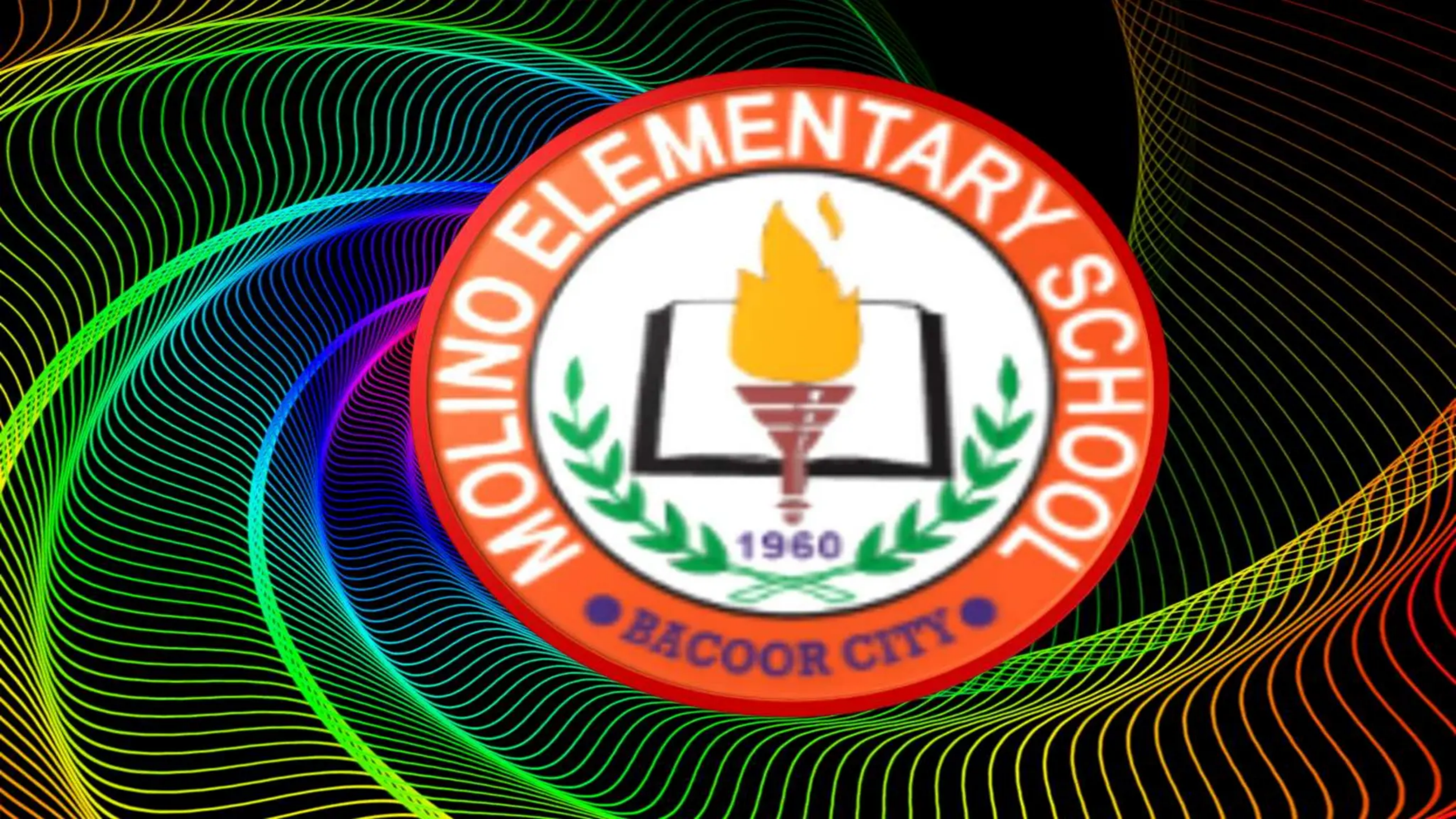

MOLINO ELEMENTARY SCHOOL
1960
BACOOR CITY

The 7E Instructional Model/ Learning Cycle

7E MODEL

1. Enumerate and explain the different stages/ phases of the 7E instructional model/ learning cycle.
2. Identify instructional strategies/ activities for each phase of the 7Es.
3. Design an instructional plan in Science using the 7E model.

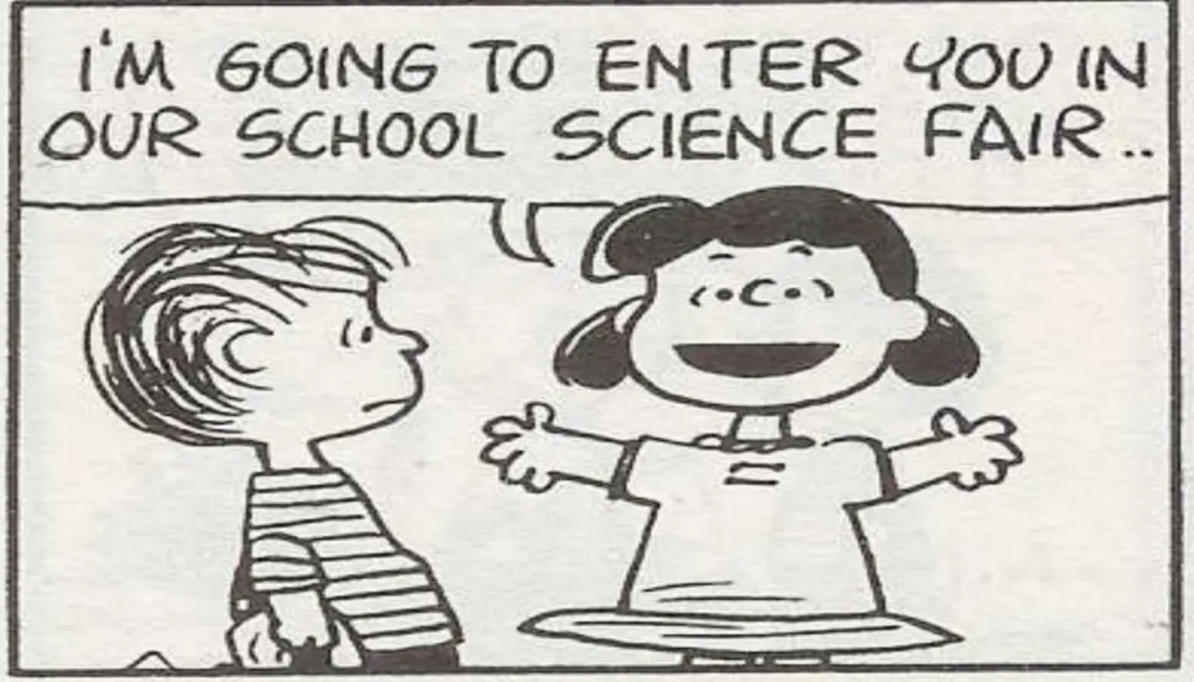
WE'RE GOING TO HAVE A SCIENCE FAIR AT SCHOOL... I'D SURE LIKE TO WIN A RIBBON..

I'VE GOT TO COME UP WITH SOME KIND OF PROJECT THAT WILL BE SO ORIGINAL AND SO DIFFERENT THAT I'LL BE CERTAIN TO WIN!

ALL THE OTHER KIDS WILL HAVE ROCKS AND BUGS AND BATTERIES AND MICE AND SEEDS AND ALL OF THAT STUFF... I'VE GOT TO THINK OF SOMETHING COMPLETELY DIFFERENT..

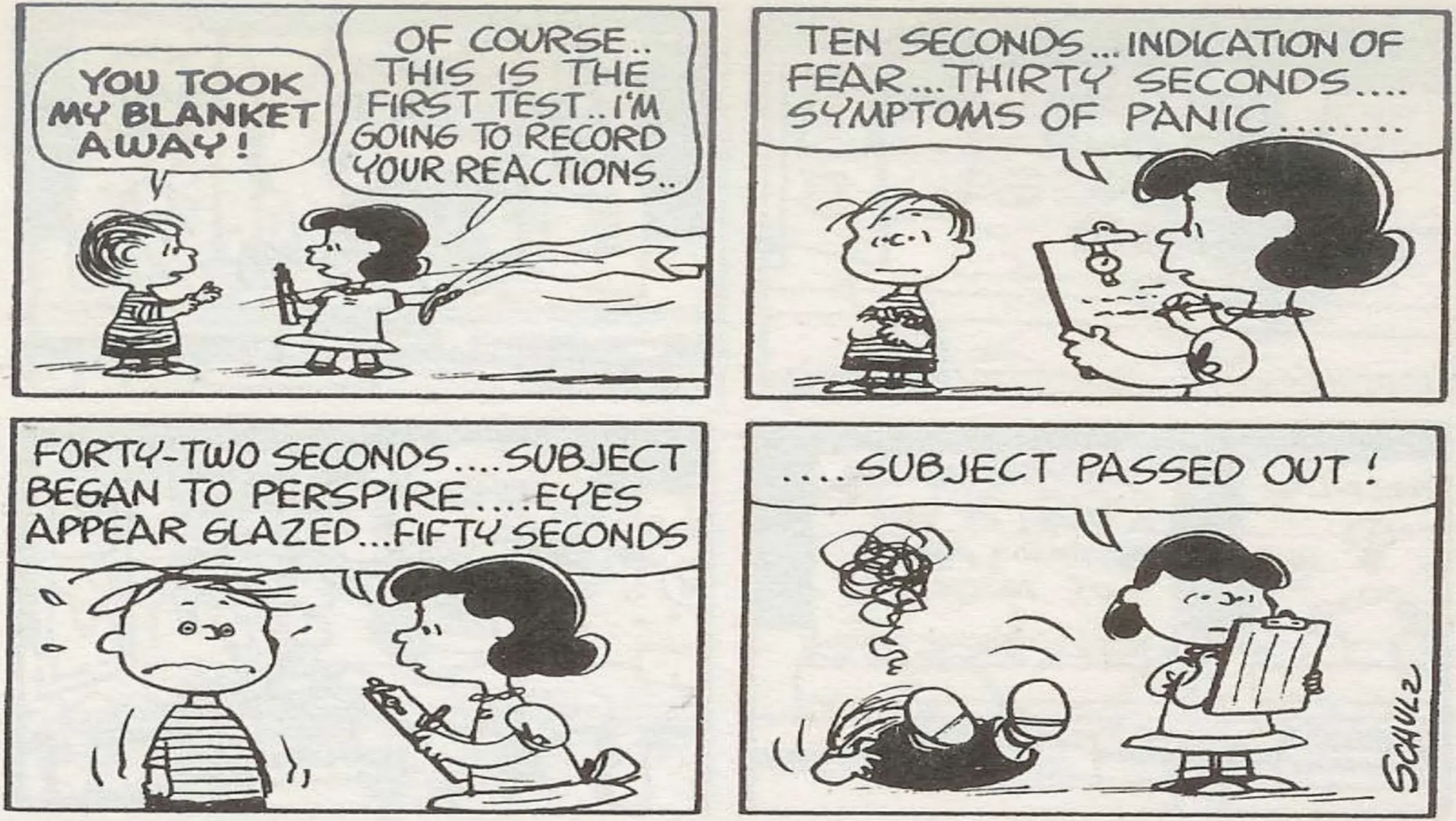
THAT'S IT!
SCHULZ

I'M GOING TO BE WHAT?
YOU'RE GOING TO BE MY SCIENCE PROJECT!

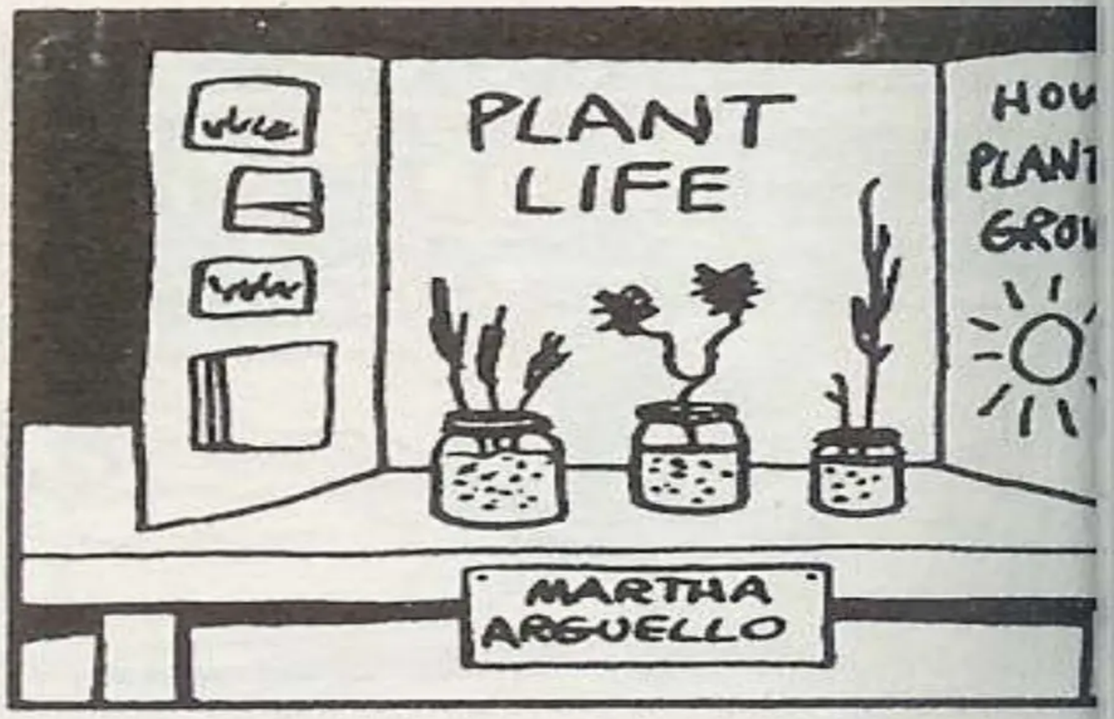

I'M GOING TO ENTER YOU IN OUR SCHOOL SCIENCE FAIR..

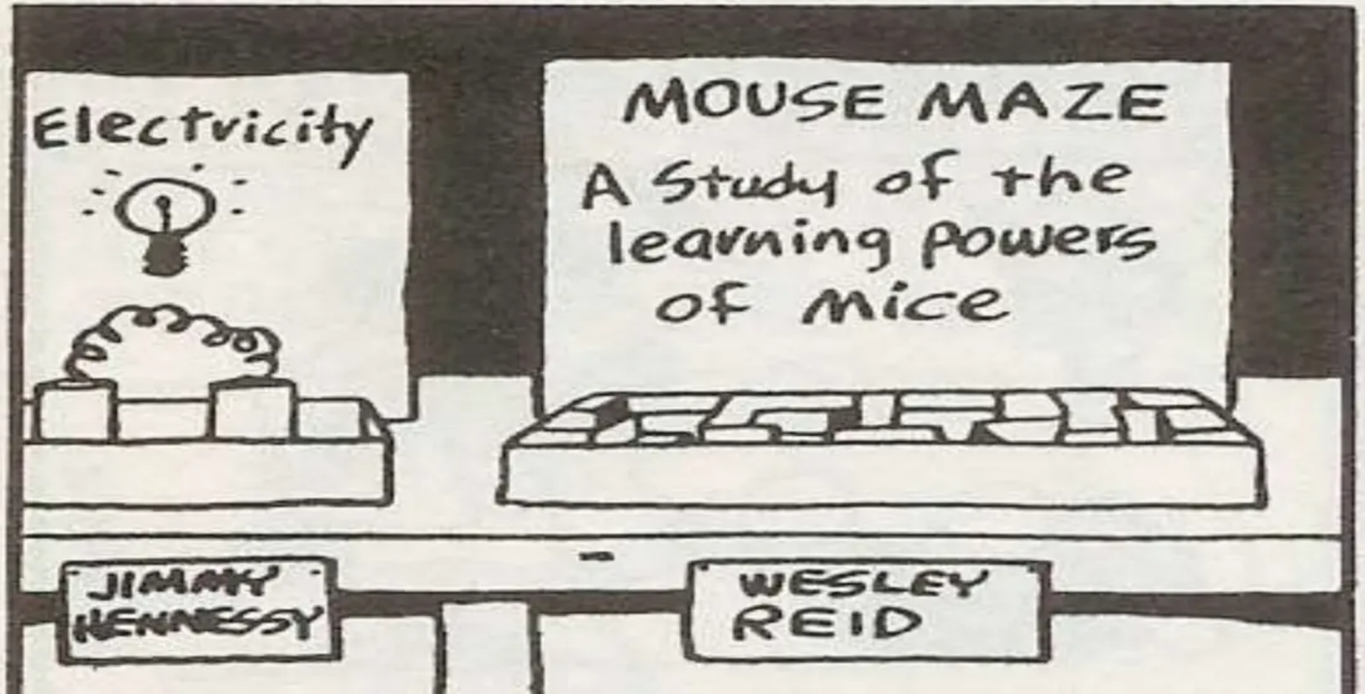
I'M GOING TO MAKE A SERIES OF TESTS WITH YOU AND THAT STUPID BLANKET TO SEE WHY IT BRINGS YOU SECURITY..

SUDDENLY I FEEL VERY INSECURE!
SCHULZ


YOU TOOK MY BLANKET AWAY!
OF COURSE.. THIS IS THE FIRST TEST..I'M GOING TO RECORD YOUR REACTIONS..
TEN SECONDS...INDICATION OF FEAR...THIRTY SECONDS.... SYMPTOMS OF PANIC.....
FORTY-TWO SECONDS....SUBJECT BEGAN TO PERSPIRE....EYES APPEAR GLAZED...FIFTY SECONDS
....SUBJECT PASSED OUT!
SCHULZ

WELCOME TO OUR SCIENCE FAIR


PLANT LIFE
MARTHA ARGUELLO


Electvicity
MOUSE MAZE
A Study of the learning powers of mice
JIMMY HENNESSY
WESLEY REID

SIGH!
THE SECURITY BLANKET
1
LUCY VAN PELT

YOU KNOW, LUCY.. YOU REALLY PUZZLE ME...

WHAT SORT OF PERSON ARE Y
WHO WOULD ENTER HER OWI
BROTHER IN A SCIENCE FAIR

I WON, DIDN'T I?

THAT'S ALWAYS BEEN A PROBL
WITH ME ...I'VE NEVER KNOWN
HOW TO ARGUE WITH SUCCE

Questions:

- **What type of presentation is shown by the 4 slides ?**
- **What should be the topic of the lesson if this comic strip is presented?**
- **What is the main idea shown in the situation**
- **Why is it presented, correlate it to your teaching learning activity as a teacher.**

What is 7E Instructional Model/ Learning Cycle?

How does the 7E model promote Active, Collaborative, Inquiry-Based Learning?

In a classroom setting, a teacher can facilitate the inquiry process by guiding students through the stages of questioning, investigation, and analysis.

Teachers can introduce students to the scientific method and guide them in formulating hypotheses, identifying variables, and planning their experiments.

Teachers can also provide resources and support to ensure students conduct valid and reliable experiments.

Teachers can encourage students to think critically by asking thought-provoking questions, challenging assumptions, and evaluating the validity of their findings.

Teachers can provide opportunities for students to present their projects to their peers, encouraging them to develop presentation skills and effectively communicate their research methods, results, and conclusions.

S

E

V

E

n

- is an instructional model/ learning cycle
- describes a teaching sequence that helps students build their own understanding based on their prior knowledge or experience.
- promotes scientific inquiry and constructivism

S E V E n

ELICIT

ENGAGE

EXPLORE

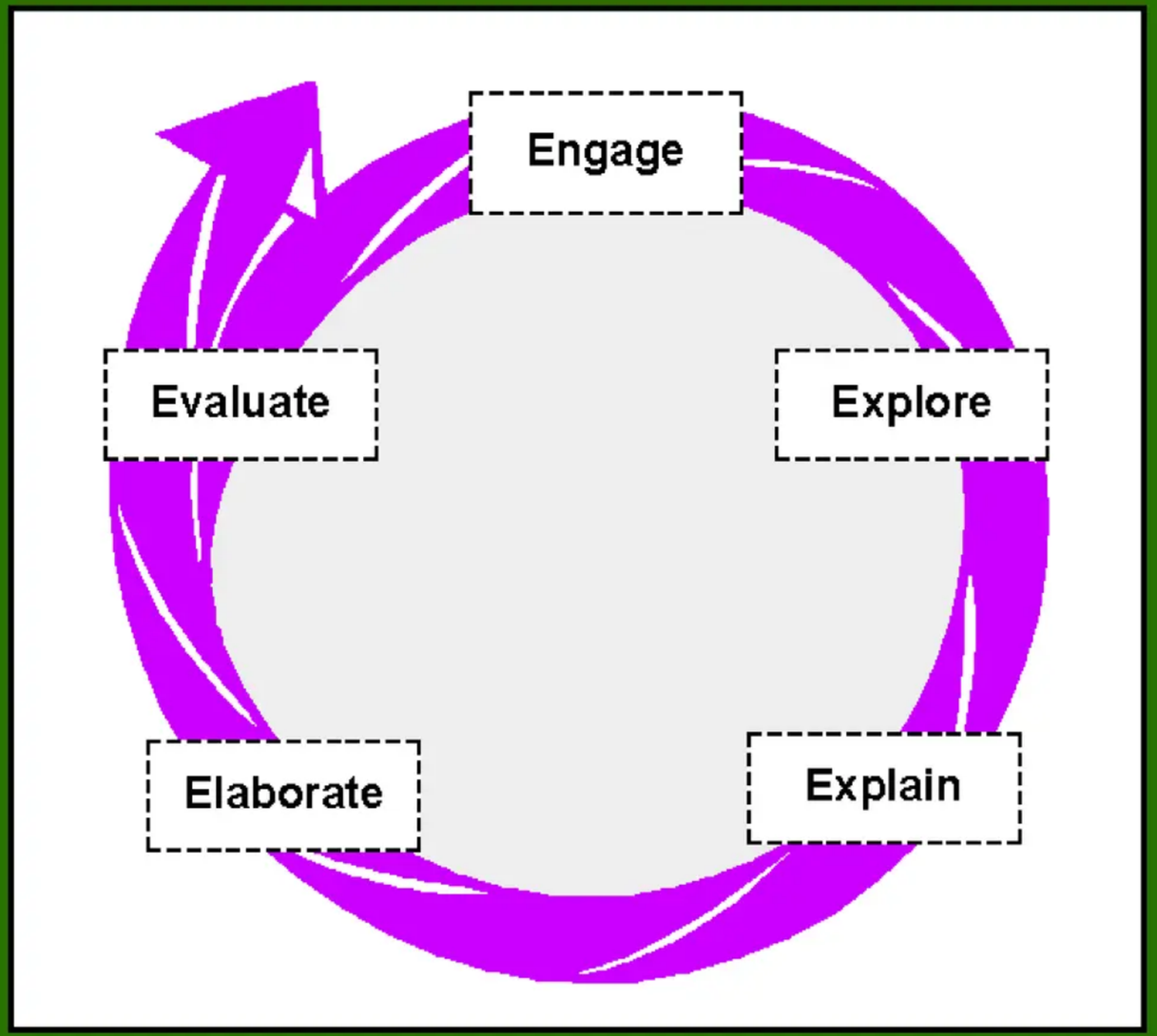
EXPLAIN

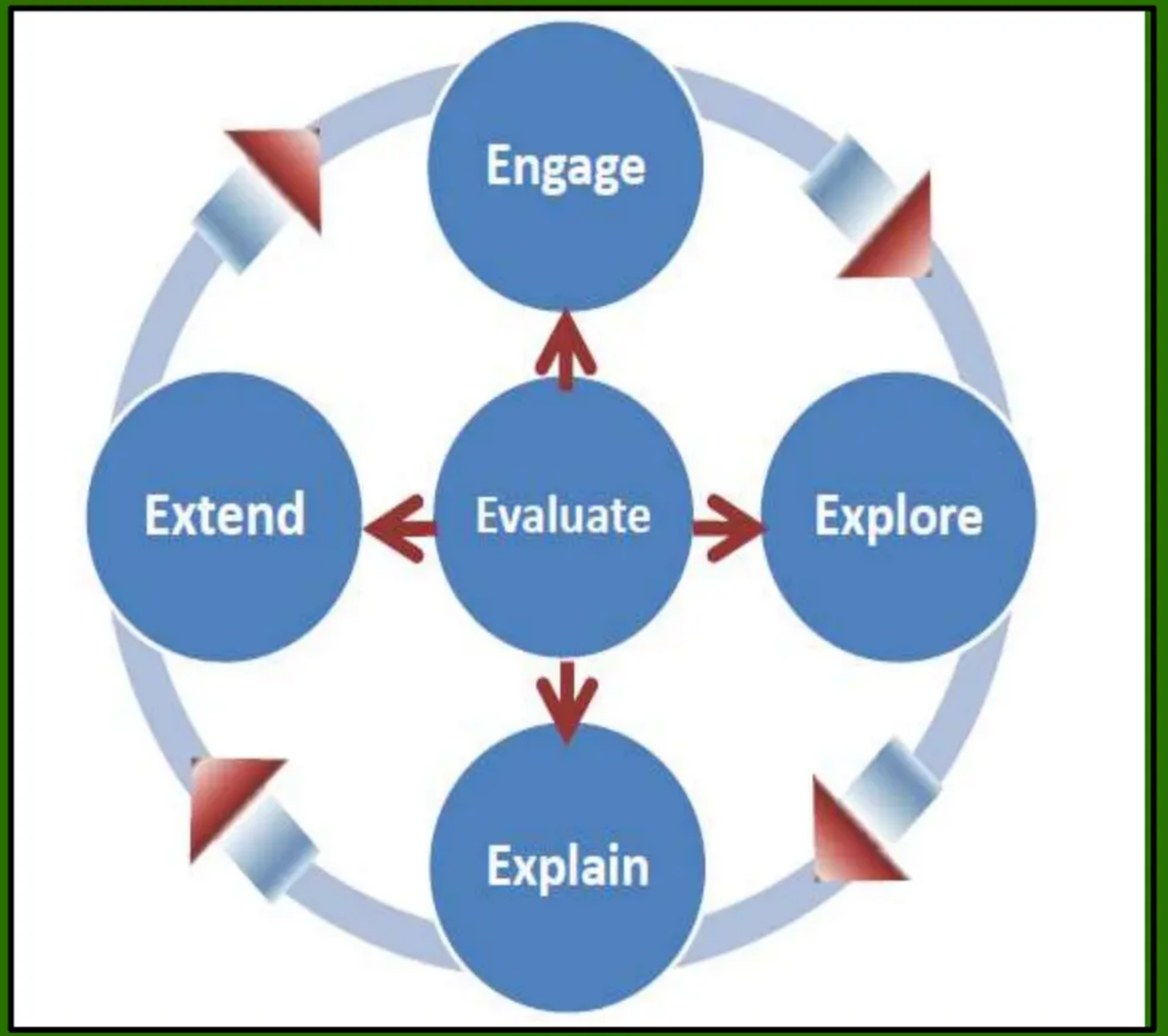
ELABORATE

Evaluate

EXTEND

The 5E Instructional Model/ Learning Cycle





5E to 7E

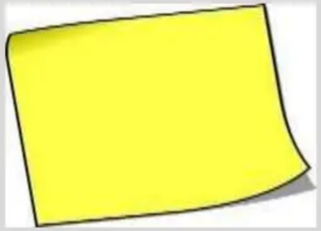
5E	7E
Engage	Elicit
	Engage
Explore	Explore
Explain	Explain
Elaborate	Elaborate
	Evaluate
Evaluate	Extend

ELICIT

- *Elicits prior understandings*
- Extracts or draws attention to prior understandings and knowledge
- Assists in transferring knowledge
- Framing a “what do you think” question

Elicit

Show me boards



Post its

Traffic light confidence at start

Think pair share

Games

What I Know	What I want to know	What I learnt

KWL boards

Multiple choice

True or false

Dominoes

Assessment FOR LEARNING

ELICIT

Examples/Suggestions for Classroom Instruction

- K of the KWL Chart (K= Know)
- Think-Pair-Share
- Draw-What-You-Know

ENGAGE

Activity which will focus student's attention, stimulate their thinking, and access prior knowledge.

- Concept Cartoon
- Concept Map
- Predict-Observe-Explain (Probex or POE)
- Demonstration
- Free Write
- Graphic Organizer
- Brainstorming
- Think-Pair-share

ENGAGE

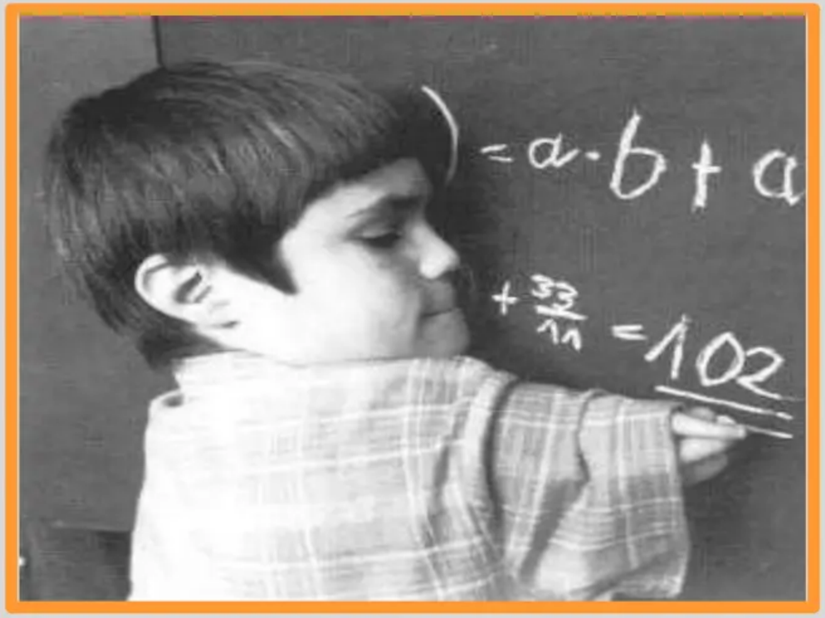
- captures students’ attention
- gets students thinking about the subject matter
- raises questions in students’ minds
- Stimulates thinking, and access prior knowledge.

Engage tasks

Video clips

Demonstration

BIG question



Stimulating image

ENGAGE

Examples/Suggestions for Classroom Instruction

- Think-Pair-Share to provide conversation opportunities
- Demonstration by teacher with written observations
- Foldables for creating visual representations of content
- Menu Choice Boards – students select optional learning
- Student created skits to explain or represent knowledge

ENGAGE

Examples/Suggestions for Classroom Instruction *(cont'd)*

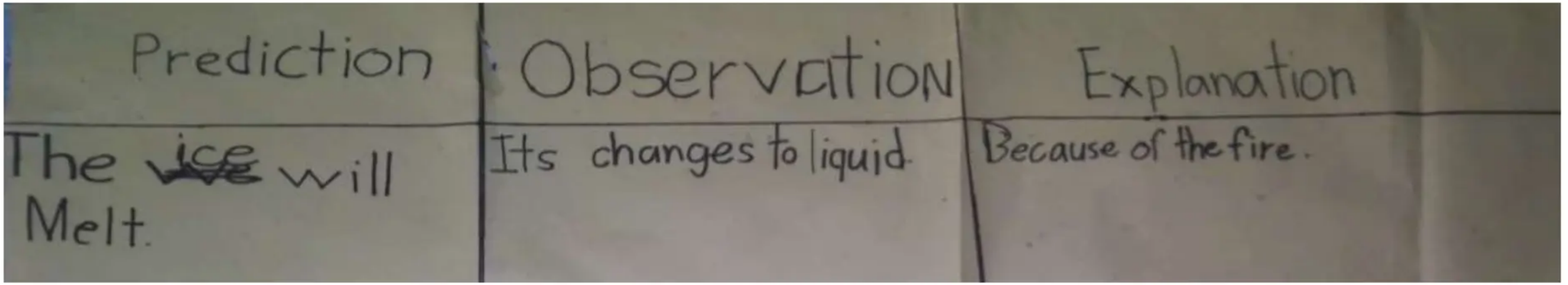
- One-to-one technologies (ie: graphing calculators, interactive white boards, interactive Websites)
- Graphic Organizers
- Inquiry-Based Learning Stations: discovery, student-centered

LC: Describe changes in properties of materials when exposed to certain conditions such as <u>temperature</u> or when mixed with other materials. (S4MT-Ig-h-6)

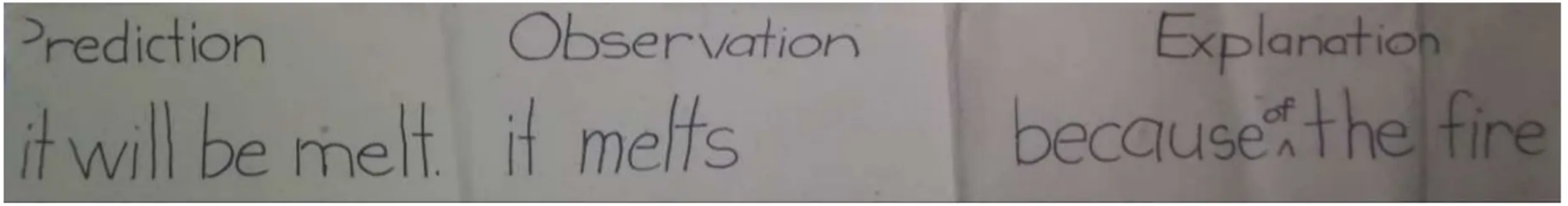
Activating learners' prior knowledge

Predict-Observe-Explain (PROBEX or POE Strategy)

Q: What do you think will happen to this solid material (ice cube) when it will be heated?

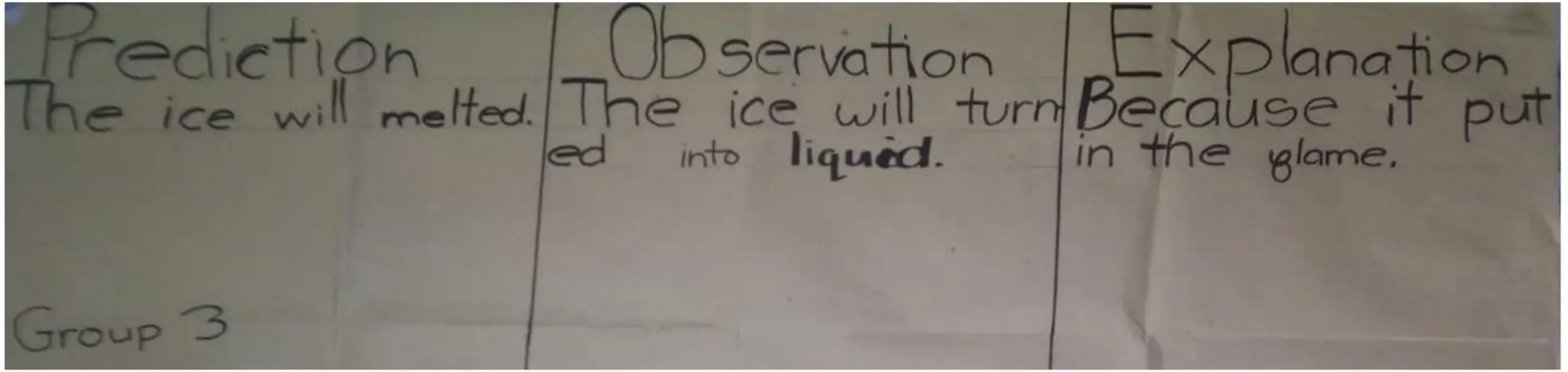


Prediction	Observation	Explanation
The ice will Melt.	Its changes to liquid.	Because of the fire.

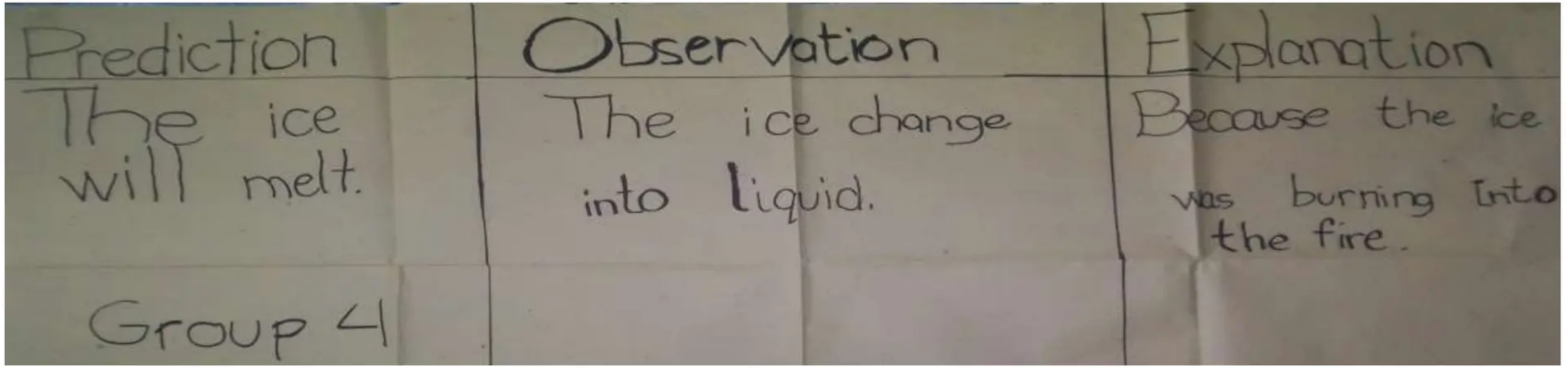


Prediction	Observation	Explanation
it will be melt.	it melts	because of the fire

Predict-Observe-Explain (PROBEX or POE Strategy)



Prediction	Observation	Explanation
The ice will melted.	The ice will turned into liquid.	Because it put in the flame.
Group 3		



Prediction	Observation	Explanation
The ice will melt.	The ice change into liquid.	Because the ice was burning into the fire.
Group 4		

Engaging learners to a scientifically-oriented question

EXPLORE

Activity which gives students time to think and investigate/ test/ make decisions/ problem solve, and collect data/ information.

- Experimentation
- Investigation/ Inquiry
- Research Authentic Resources to Collect Information
- Solve a Problem
- Construct a Model

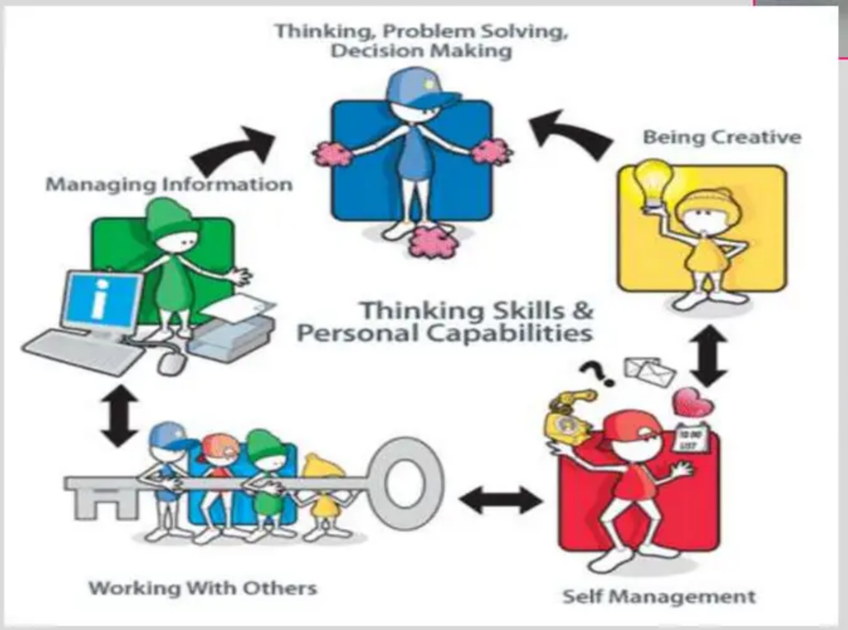
EXPLORE

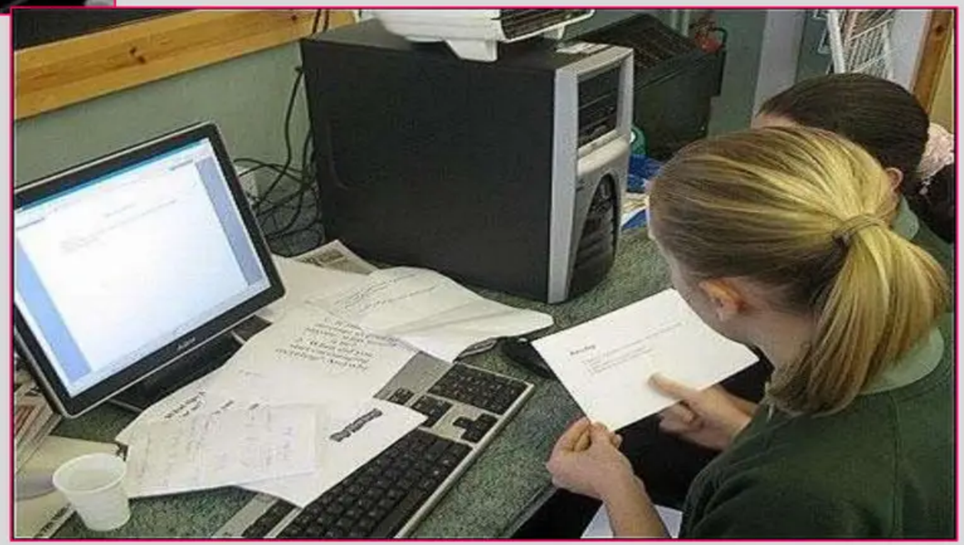
- provides an opportunity for students to observe, record data, isolate variables, design and plan experiments, create graphs, interpret results, develop hypotheses, and organize their findings.
- Teachers may frame questions, suggest approaches, provide feedback, and assess understandings.

Explore

VAK

Practical activities





Research activities

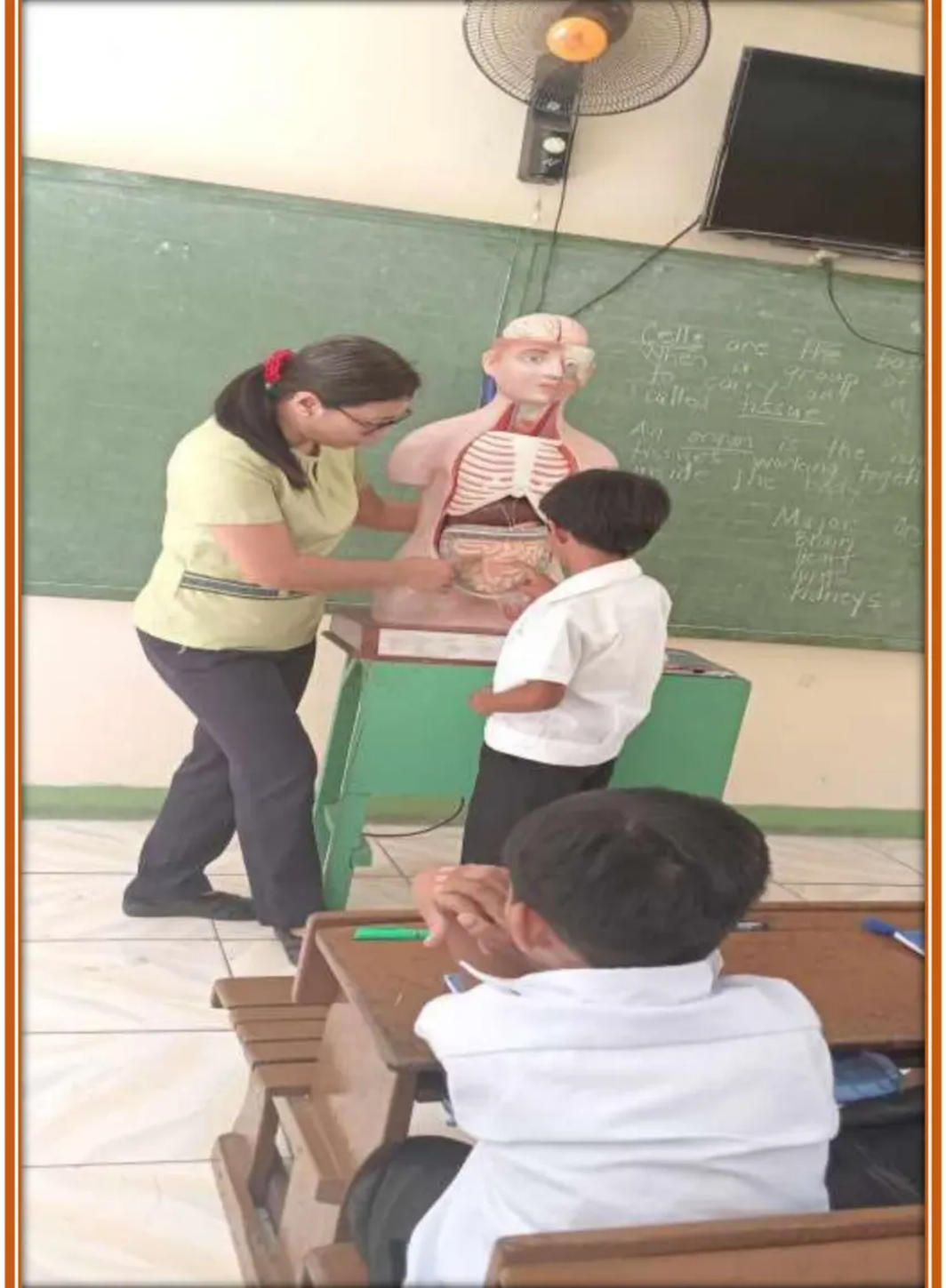
Information retrieval

EXPLORE

Examples/Suggestions for Classroom Instruction

- Students plan and carry out investigations
- Students analyze and interpret data
- Students make predictions from demonstrations
- Question prompts by the teacher
- Data collection during lab activities and in science experiments
- Cooperative group learning activities
- Jigsaw groups where student become group experts and then travel to other groups to share their specific components
- Student created graphs

Bones-
Muscles -


Cells are the
When a group
called tissue
An organ is the
Major
kidneys

muscles
bones

P.F

EXPLAIN

Activity which allows students to analyze their exploration. Student's understanding is clarified and modified through a reflective activity.

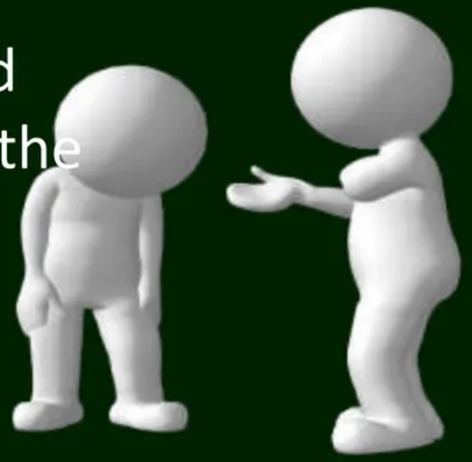
- Student Analysis & Explanation
- Supporting Ideas with Evidence
- Structured Questioning
- Reading/ Reporting and Discussion
- Teacher Explanation
- Thinking Skill Activities: compare, classify, error analysis

EXPLAIN

- Students are introduced to models, laws, and theories during the *explain* phase of the learning cycle.

- Students summarize results in terms of these new theories and models.

- The teacher guides students toward coherent and consistent generalizations, helps students with distinct scientific vocabulary, and provides questions that help students use this vocabulary to explain the results of their explorations.

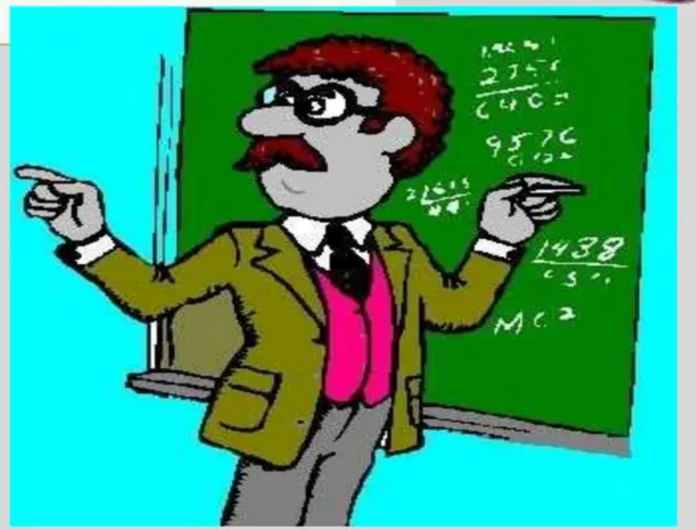


Explain

Teacher's Role	Student's Role
• Formally provides definitions, explanations, and new labels • Uses students' previous experiences as a base for explaining concepts (*revisit hands-on activities) • Encourages students to explain concepts and definitions in their own words	• Listens officially to others explanations and questions them • Makes connections to prior knowledge • Records observations and explanations • Explains possible solutions to others

Assessment

FOR LEARNING

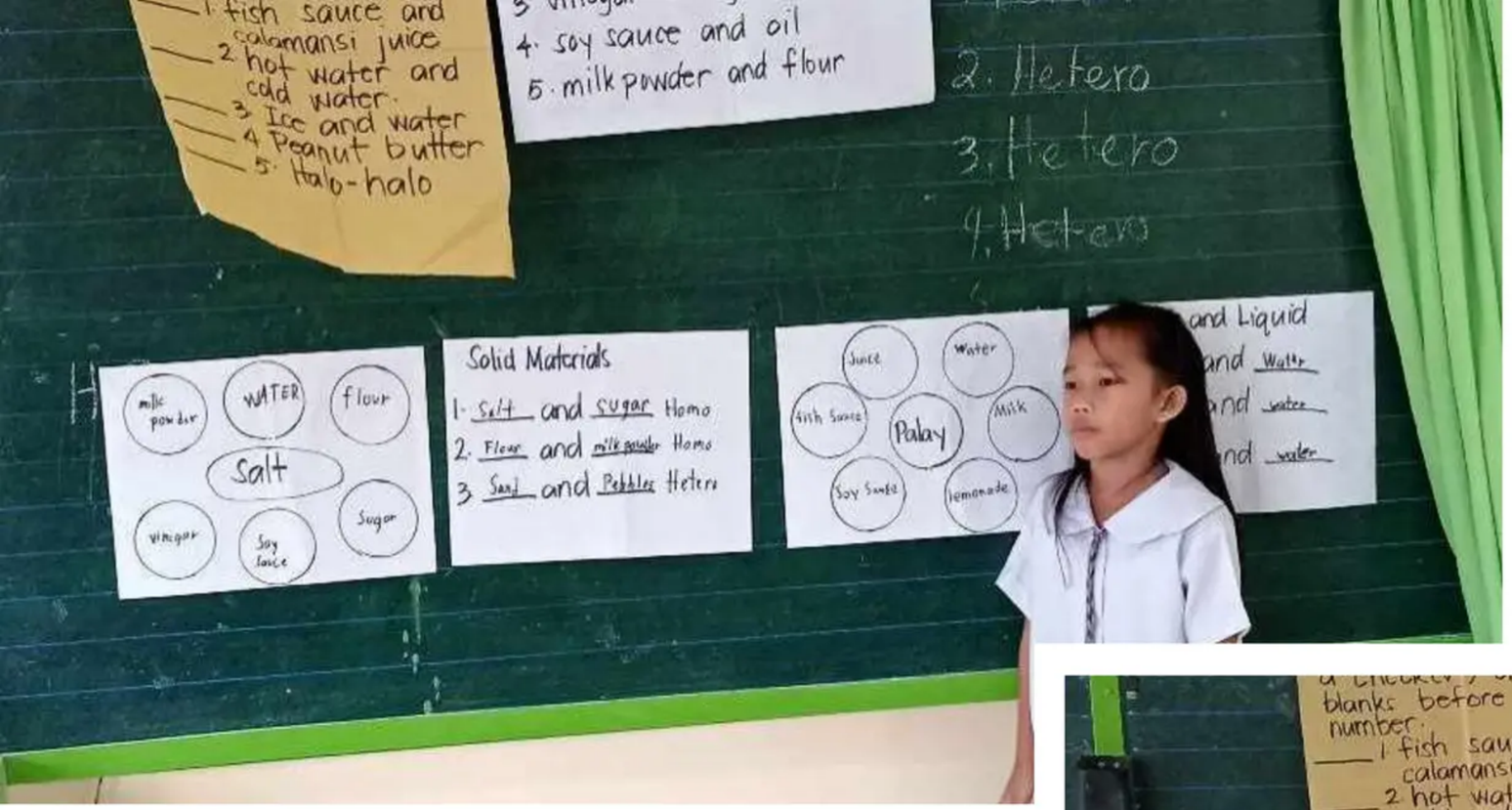


EXPLAIN

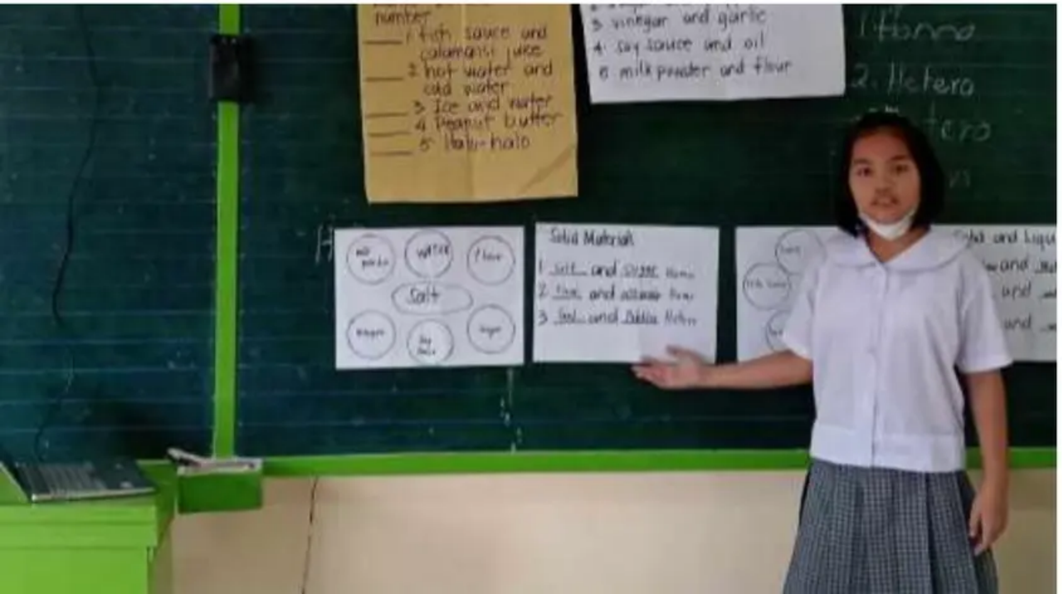
Examples/Suggestions for Classroom Instruction

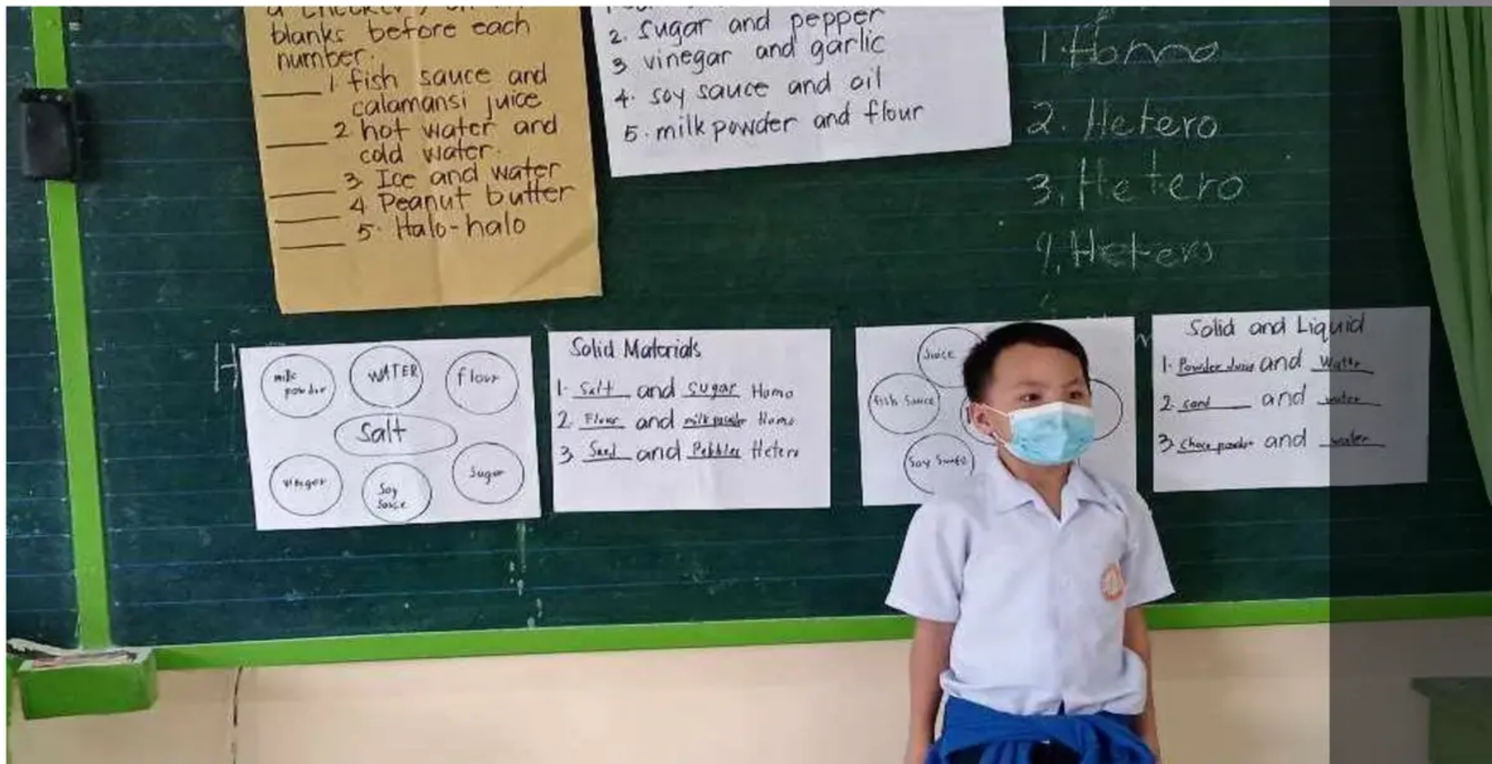
- Students construct explanations and design solutions
- Engage in arguments from evidence
- Obtain, evaluate and communicate information
- Expository Writing (ie: What happened during the lab? Why did this happen? Will the results always be the same?
- Peer-to-Peer verbal review and clarification
- Oral presentation of lab results
- Oral presentation of project
- Log Book
- Journal writings and reflections
- Science Projects with abstracts and summaries

of Group's
Output
diseases
unhealthy
strong
lifestyle
habits
healthy
MAJOR ORGAN
CONNECTIONS
MAJOR ORGAN
Effect
Cause
Taking good care of your Organs
Effect
11-17-23
B-21
4-76


1. fish sauce and calamansi juice
2 hot water and cold water.
3 Ice and water
4 Peanut butter
5. Halo-halo
4. soy sauce and oil
5. milk powder and flour
2. Hetero
3. Hetero
4. Hetero
milk powder
WATER
flour
Salt
vinegar
Soy Sauce
Sugar
Solid Materials
1. Salt and Sugar Homo
2. Flour and milk powder Homo
3. Sand and Pebbles Hetero
Juice
Water
fish Sauce
Palay
Milk
Soy Sauce
lemonade
and Liquid
and Water
and water
and water

5. milk powder and flour
Salt


1 fish sauce and calamansi juice
2 hot water and cold water.
3 Ice and water
4 Peanut butter
5 Halo-halo
5 vinegar and garlic
4 soy sauce and oil
5 milk powder and flour
Solid Materials
Salt
1. Salt and Sugar Homo
2. Flour and milk powder Homo
3. Sand and Pebbles Hetero


blanks before each number.
1 fish sauce and calamansi juice
2 hot water and cold water.
3 Ice and water
4 Peanut butter
5 Halo-halo
2. Sugar and pepper
3 vinegar and garlic
4 soy sauce and oil
5 milk powder and flour
1. Homo
2. Hetero
3. Hetero
4. Hetero
milk powder
WATER
flour
Salt
vinegar
Soy Sauce
Sugar
Solid Materials
1. Salt and Sugar Homo
2. Flour and milk powder Homo
3. Sand and Pebbles Hetero
Juice
fish Sauce
Soy Sauce
Solid and Liquid
1. Powder Juice and Water
2. sand and water
3. Choco powder and water

ELABORATE

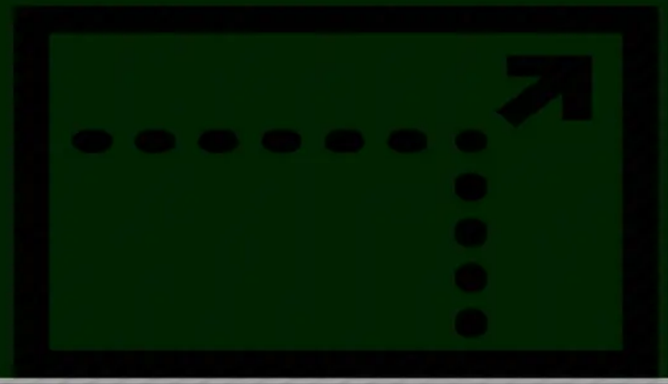
Activity which expands and solidifies student thinking and/or applies it to a real-world situation.

- Enrichment/ Reinforcement
- Problem Solving
- Verification activity
- Decision Making
- Experimental Inquiry
- Thinking Skill Activity
- Video-based activity

Honesty is the best policy

ELABORATE

- Transfer of learning
- Transfer one concept to another
- Transfer subject to subject
- Application to a new context
- Refers to activities that build on, extend and refine, and/or require the application and use of the scientific concepts and vocabulary in new situations. Activity is designed to increase the depth and breadth of student understanding.



Elaborate

Role play

Application of knowledge

Level assessed task

Conclusions or reports

Demonstrates learning

Assessment
FOR LEARNING

ELABORATE

Examples/Suggestions for Classroom Instruction

- Assessments which include questions related to labs and require application of new knowledge
- Science Olympiad Events
- Students design a product which applies findings to authentic situations
- Students create performance tasks
- Students design real-life solutions to existing problems based on new knowledge
- Book study on nonfiction books related to the specific content area

EVALUATE

Activity which allows the teacher to assess student performance and/or understandings of concepts, skills, processes, and applications.

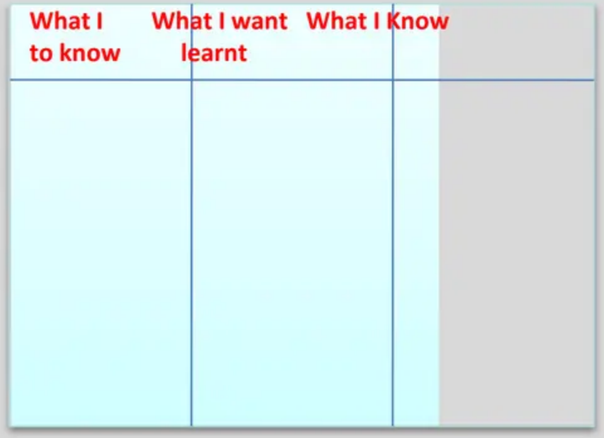
- Any of the Previous Activities
- Develop a Scoring Tool or Rubric
- Performance Assessment
- Produce a Product
- Journal Entry
- Portfolio

EVALUATE

- Formative
- Summative
- Informal
- Formal
- Provides an opportunity for learners to assess their own understanding and be able to demonstrate the depth and breadth of that understanding to others, including the teacher. Also, may allow the teacher to assess student performance and/or understandings of concepts, skills, applications

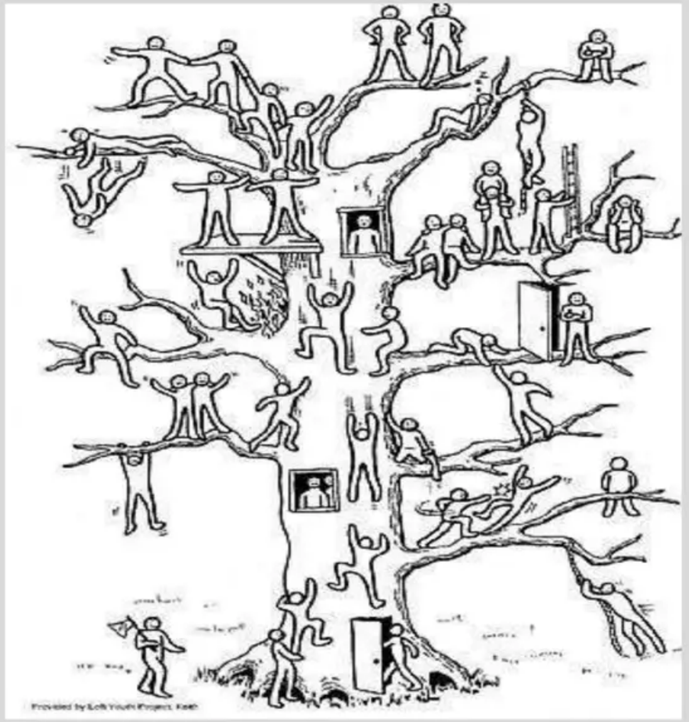
Assessment
FOR LEARNING

Evaluate tasks



KWL boards

Traffic light confidence

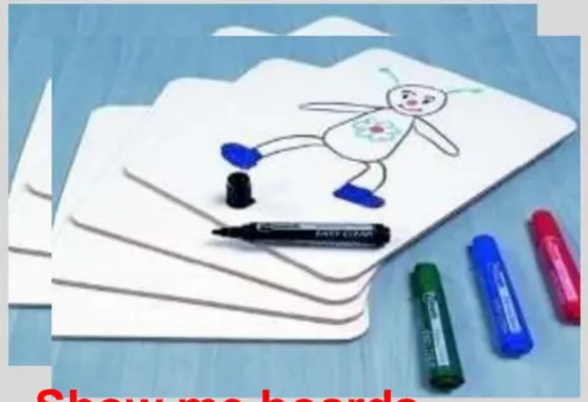


Men in the tree

Learning journals

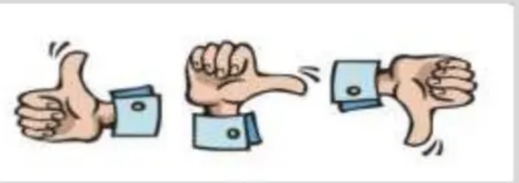
Self and peer assessment

Target setting



Show me boards

Thumbs



Learning Pyramid

Mind map

EVALUATE

Examples/Suggestions for Classroom Instruction

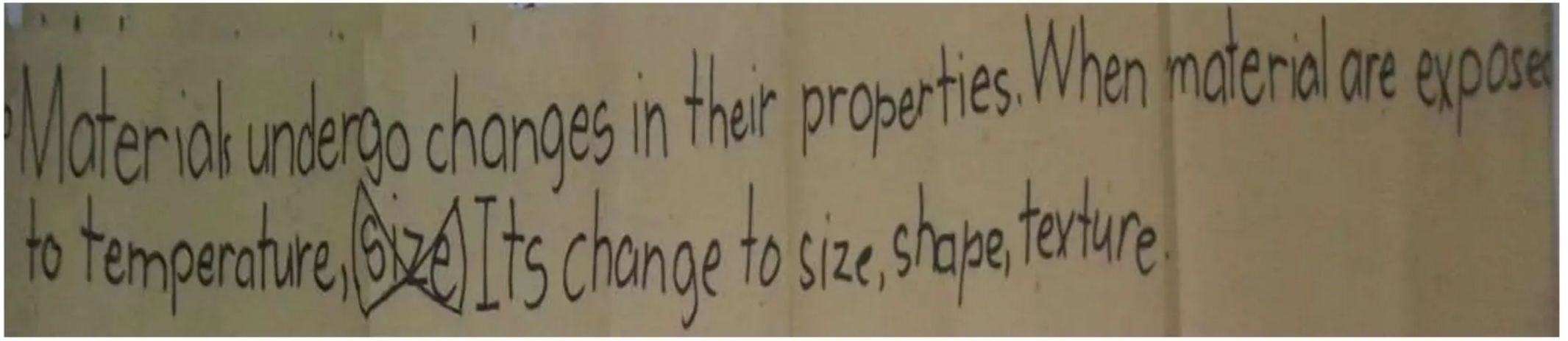
- Student converse during learning activities for student and teacher commentary/feedback
- Ticket-Out-The-Door
- Rubrics
- Self-Assessment

- **Activity-embedded assessment**

  (The results of the activity, "*What Happens to the Materials when Heated and when Cooled?*" were taken as formative assessment.)

- **Constructed-response test (open-ended question)** was also given.

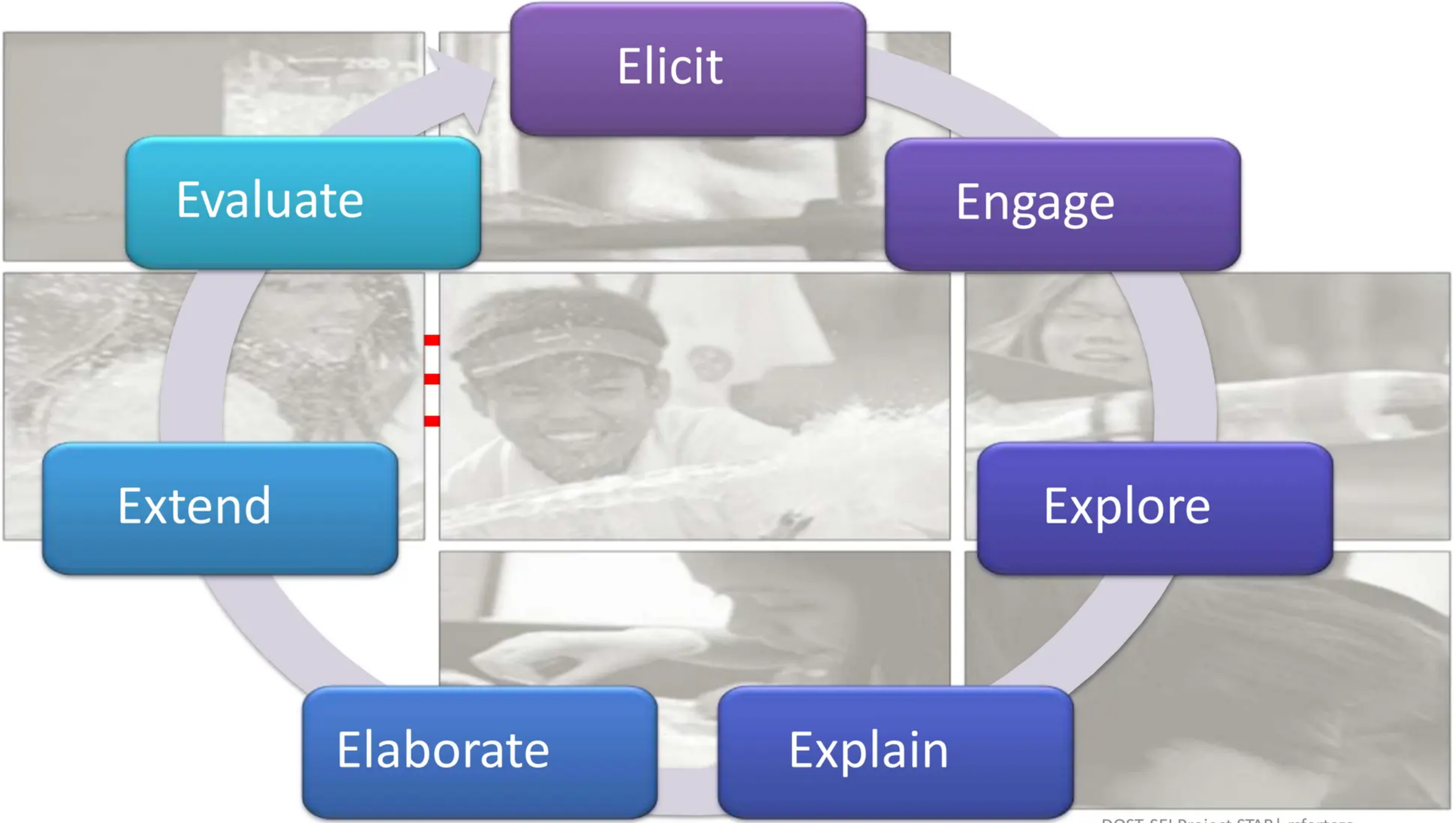
  "What happens to the materials when exposed to temperature?



EXTEND

- encouraging the students to apply or extend the concepts and skills in new situations

- Students make connections not just in the subject/ideas studied but also beyond it.
- They are able to apply ideas/generalise and transfer principles.
- This should also include self-reflection and evaluation from the student.


Elicit
Engage
Explore
Explain
Elaborate
Extend
Evaluate

How else can you further improve your competence in teaching Science

APPLICATION

Design an instructional plan in Science using the 7E model.

45 minutes

APPLICATION

7E Instructional Plan in Science

I. Objectives:
   A. Content Standards
   B. Performance Standards
   C. Enabling Competencies
   D. Learning Competencies
   E. Sub-task Learning Competencies

II. Content
   A. Topic
   B. Process Skills
   C. Value Integration
   D. Curriculum Integration

III. Learning Resources:
   A. Teacher's Guide
   B. Learner's Materials
   C. Additional Materials from LR portal
   D. Other Learning Materials

APPLICATION

7E Instructional Plan in Science

IV. Procedures

Preliminaries

Classroom Routines

Science Vocabulary

A. Elicit

B. Engage

C. Explore

D. Explain

E. Elaborate

F. Evaluate

G. Extend

“It is what teachers think,
what teachers do,
and what teachers are at the level of the classroom
that ultimately shapes the kind of learning
that young people get.”

- *Andy Hargreaves and*
Michael Fullan-

Thank you

Presented by:

JENNY B. BAUTISTA

Science Teacher

~~Seatbelt Lesson using the~~ 7E model

Elicit prior understandings

- Students are asked, “Suppose you had to design seat belts for a race car traveling at high speeds. How would they be different from the ones available on passenger cars?” The students are required to write a brief response to this “What do you think?” question in their logs and then share with the person sitting next to them. The class then listens to some of the responses. This requires a few minutes of class time.

Engage

- Students relate car accidents they have witnessed in movies or in real life.

Explore

- The first part of the exploration requires students to construct a clay figure they can sit on a cart. The cart is then crashed into a wall. The clay figure hits the wall.

Explain

- Students are given a name for their observations. Newton’s first law states, “Objects at rest stay at rest; objects in motion stay in motion unless acted upon by a force.”

~~Seatbelt Lesson using the~~ 7E model

Engage

- Students view videos of crash test dummies during automobile crashes.

Explore

- Students are asked how they could save the clay figure from injury during the crash into the wall. The suggestion that the clay figure will require a seat belt leads to another experiment. A thin wire is used as a seat belt. The students construct a seat belt from the wire and ram the cart and figure into the wall again. The wire seat belt keeps the clay figure from hitting the wall, but the wire slices halfway through the midsection.

Explain

- Students recognize that a wider seatbelt is needed. The relationship of pressure, force, and area is introduced.

Elaborate

- Students then construct better seat belts and explain their value in terms of Newton's first law and forces.

Seatbelt Lesson using the 7E model

Evaluate

- Students are asked to design a seat belt for a racing car that travels at 250 km/h. They compare their designs with actual safety belts used by NASCAR.

Extend

- Students are challenged to explore how airbags work and to compare and contrast airbags with seat belts. One of the questions explored is, “How does the airbag get triggered? Why does the airbag not inflate during a small fender-bender but does inflate when the car hits a tree?”